Enzymes

What are Enzymes?

JOMO

Characteristics of Enzymes

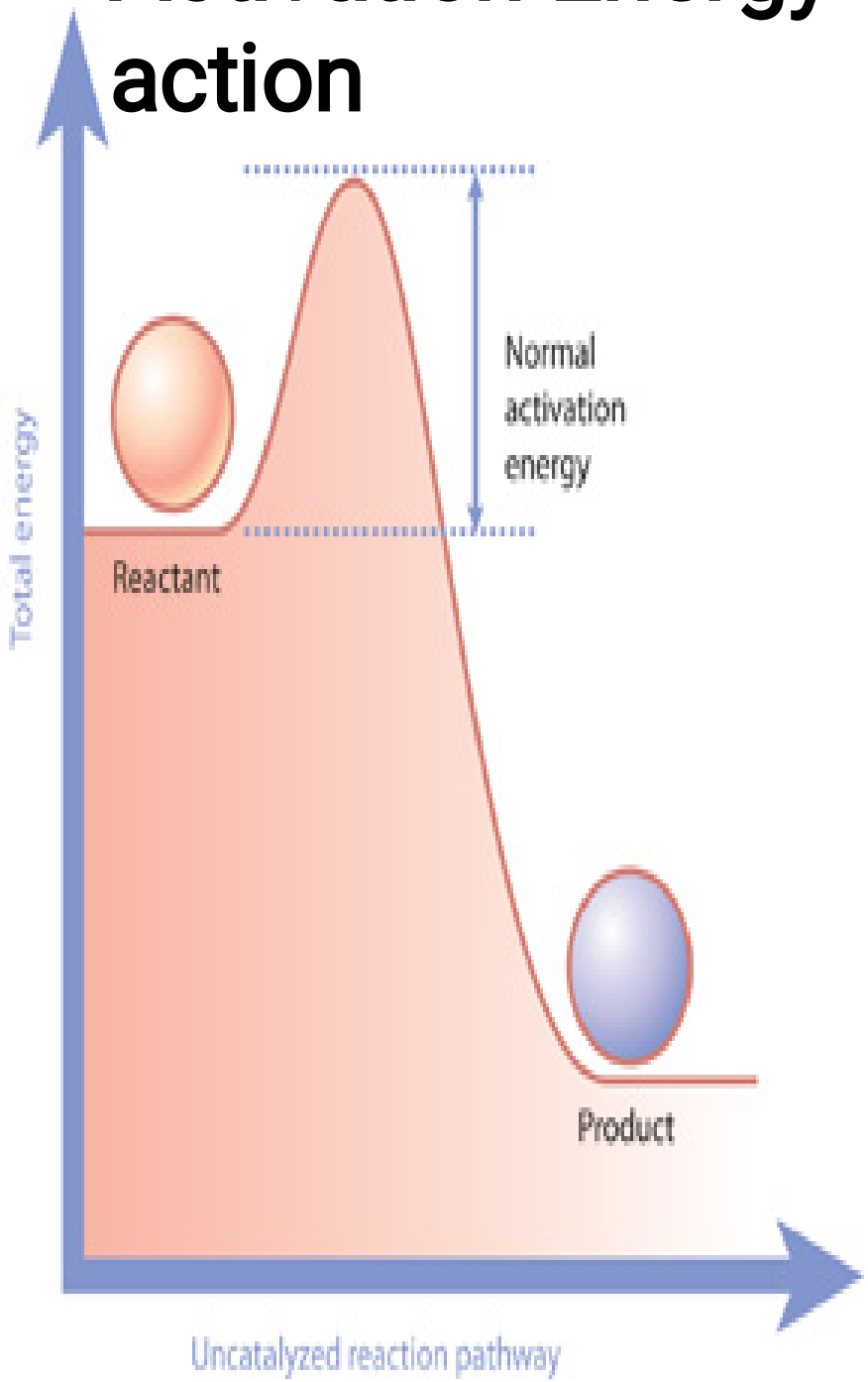
- Enzymes are **globular proteins** with an active site
- They are **biological catalyst**
- Present in **very small quantity**, i.e. Very small amount can bring about the change in a large amount of substrate. E.g. 1 molecule of enzyme **catalase** can aid the breakdown of about 600,000 molecules of H_2O_2 to H_2O and O_2 at body temp.
- Enzymes are **highly specific** to reactions they catalyse because of the shape of the active site.

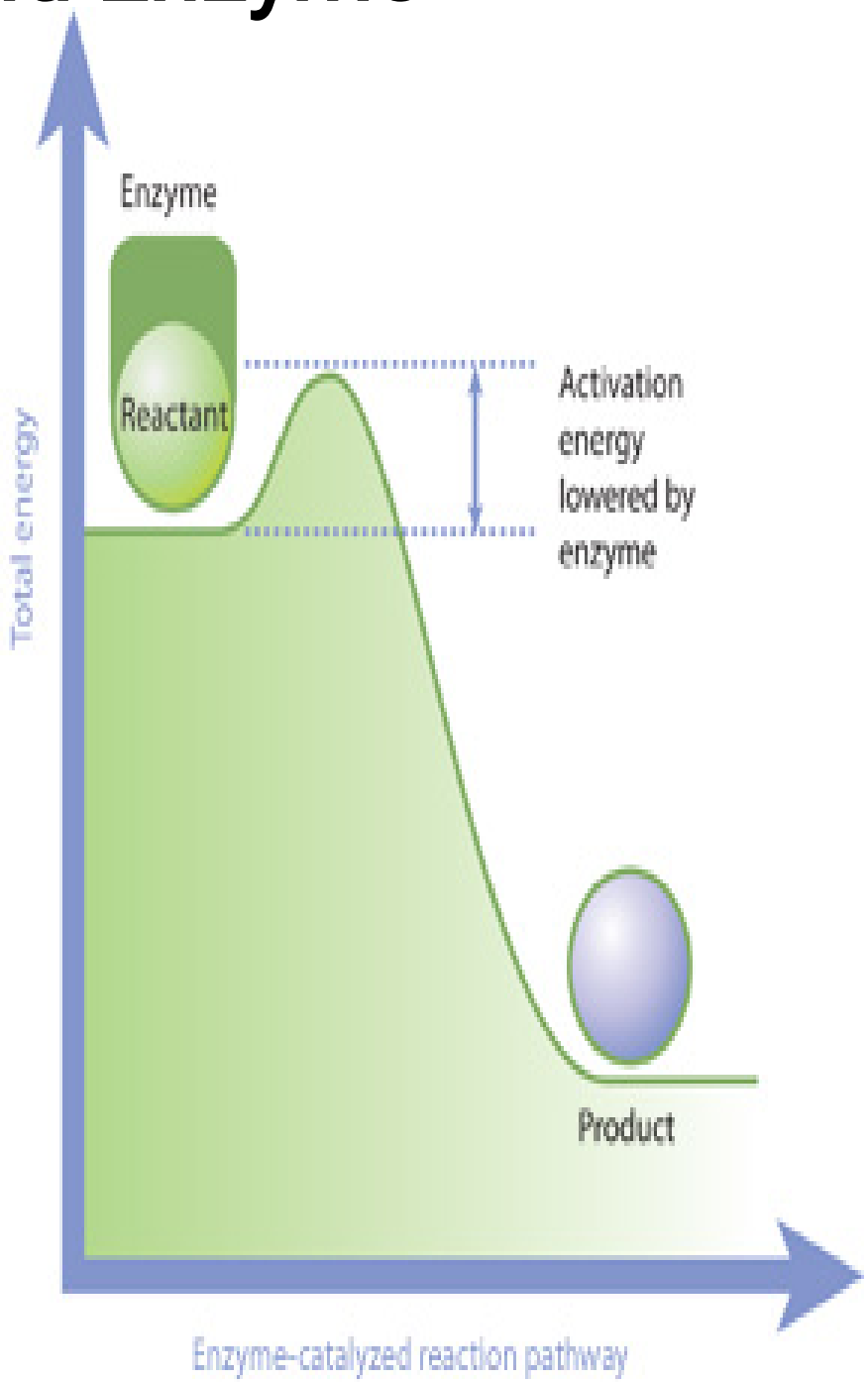
Foundation laying

- **Metabolism** is the sum total of all the reactions going on in a cell. It can be divided into:
- **Anabolism**- i.e. Anabolic reaction, involves the ***synthesis*** of molecules. It often involve condensation
- **Catabolism**- i.e. Catabolic reaction, involves the ***breakdown*** of molecules. It often involve oxidation and hydrolysis.
- All these reactions are catalysed by enzymes

- **Activation Energy** is the energy required to make substances react,i.e. destabilize existing chemical bonds & initiate chemical reaction.
- **Enzymes lower the activation energy** necessary to transform a reactant into a product.
- In the enzyme-catalyzed reaction, the enzyme binds to the reactant and facilitates its transformation into a product.
- Consequently, the enzyme-catalyzed reaction pathway has a smaller energy barrier (activation energy) to overcome before the reaction can proceed.

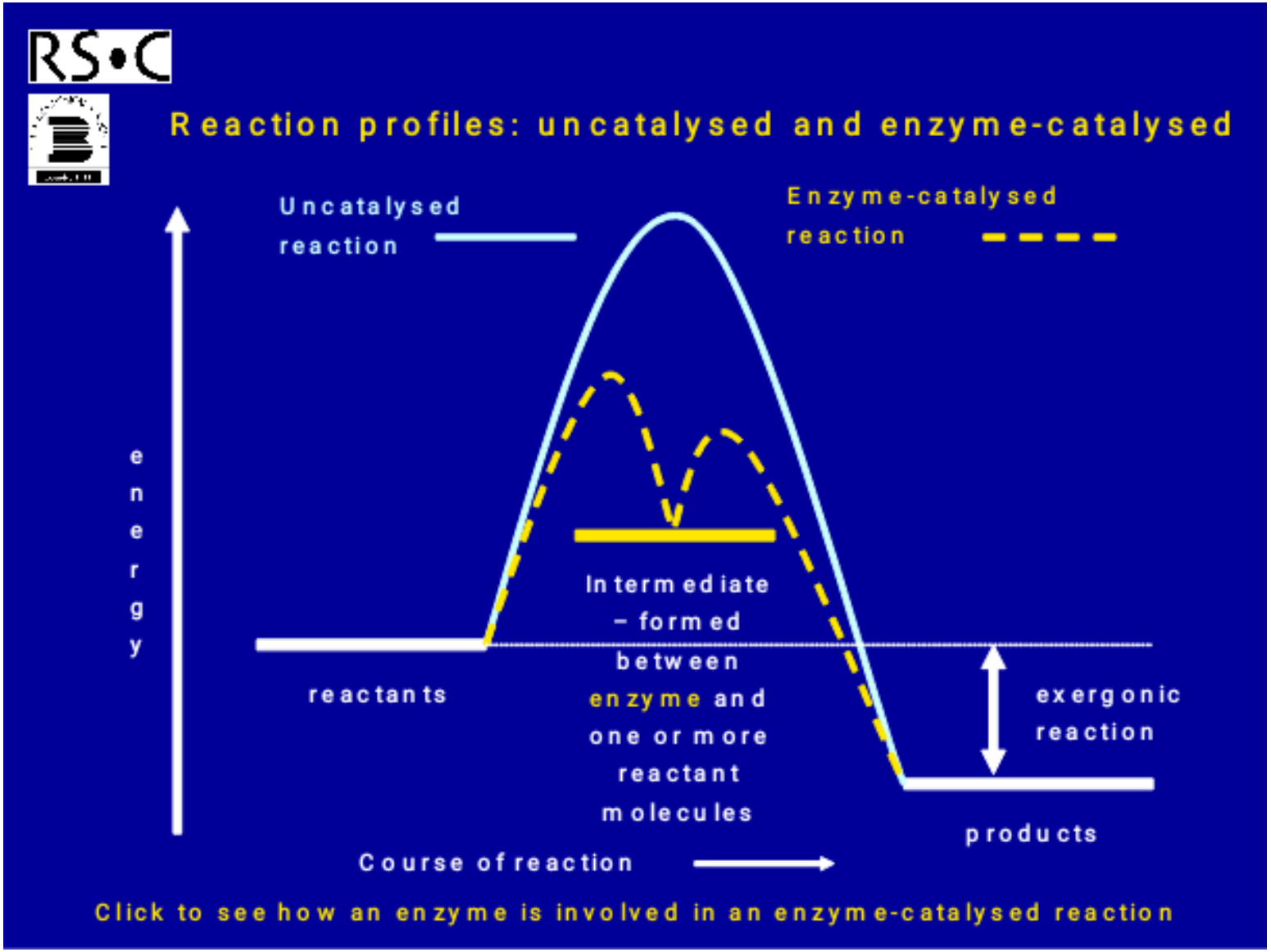
Activation Energy and Enzyme action





http://www.nature.com/scitable/topicpage/protein-function-14123348

Slide show 07



http://www.rsc.org/Education/Teachers/Resources/cfb/enzymes.htm

Simplified enzyme action sequence

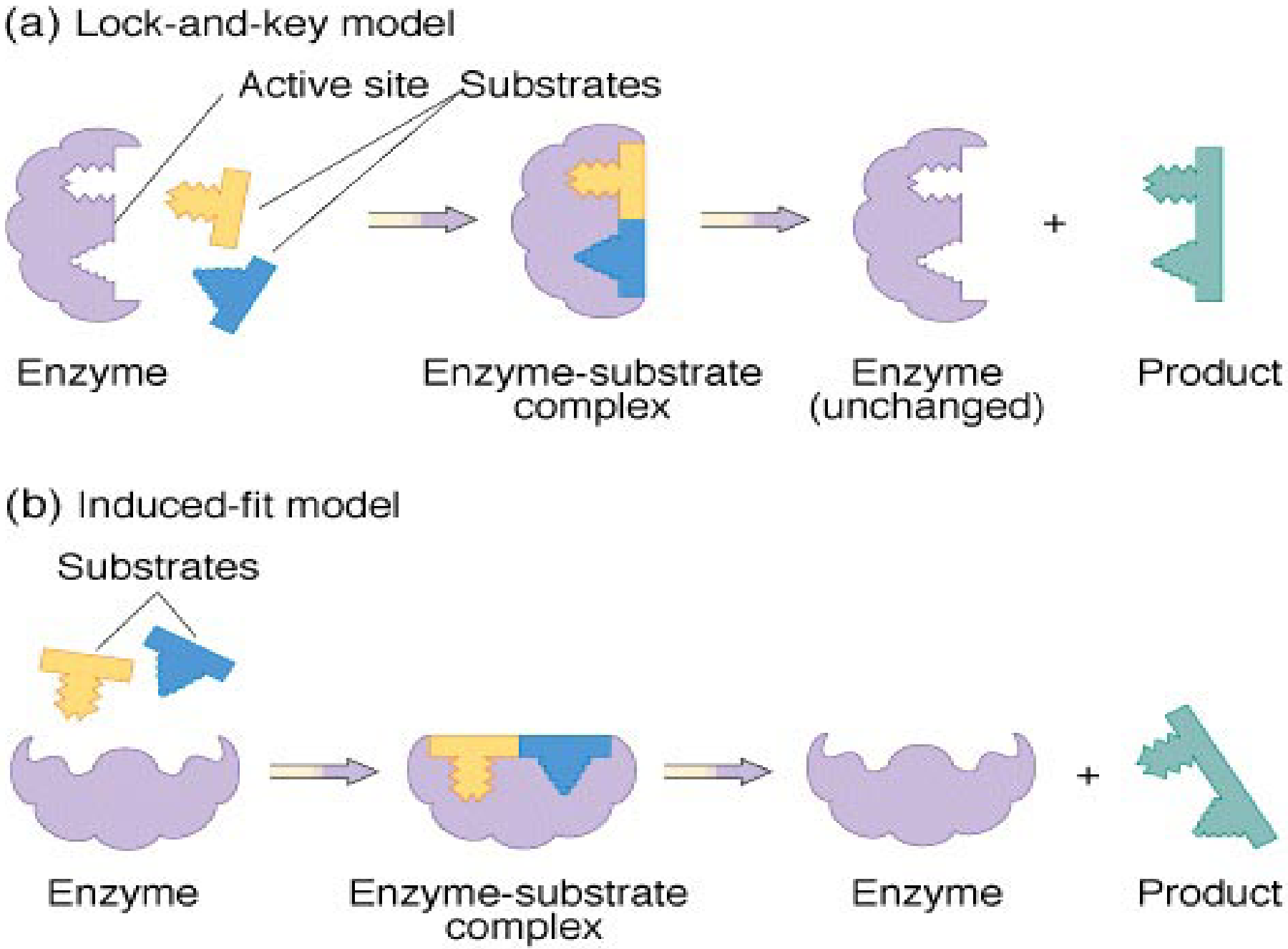
- The chemical(s) or molecule which an enzyme works on is called the **substrate**
- An enzyme combines with the substrate to form a short-lived **enzyme/substrate complex**
- The complex breaks up into **products and enzyme**
- The enzyme is unchanged and its free to interact again with more substrate

E + S <==> ES <==> E + P

Enzyme Action Hypothesis

- **Emil Fischer** (1890) –postulated the "**Lock and key" mechanism**- *rigid shapes* of Enzyme active site & Substrate structure. Confirms enzyme specificity
- **D.E. Koshland Jr** (1958)- modified lock & key by theorizing "**Induced fit" theory**- *flexible shape* of enzyme/substrate: shape of active site is modified around the substrate to form the complex. After product release, enzyme revert back to form.

Enzyme action hypothesis



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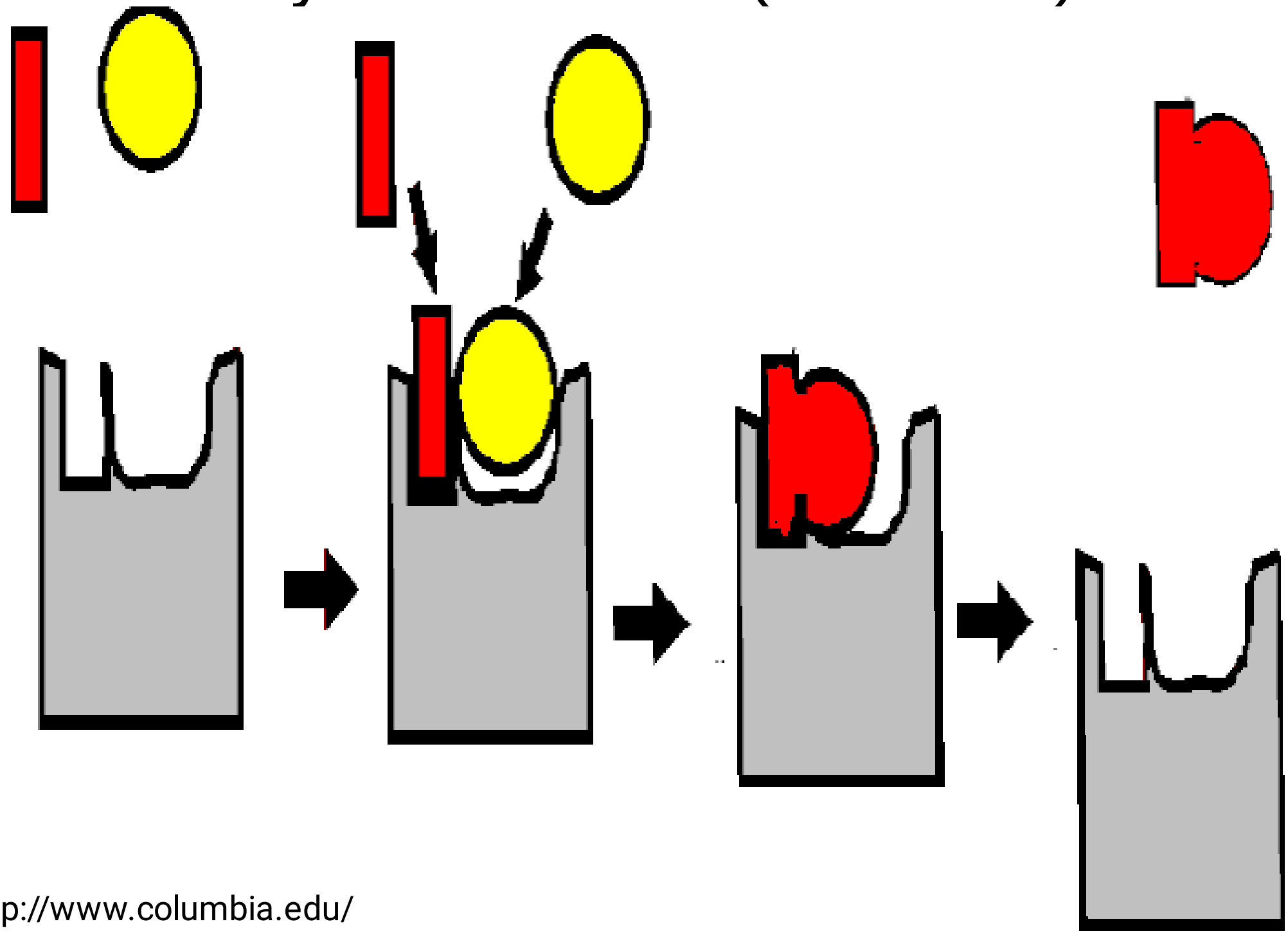
Enzyme in action (Anabolic)



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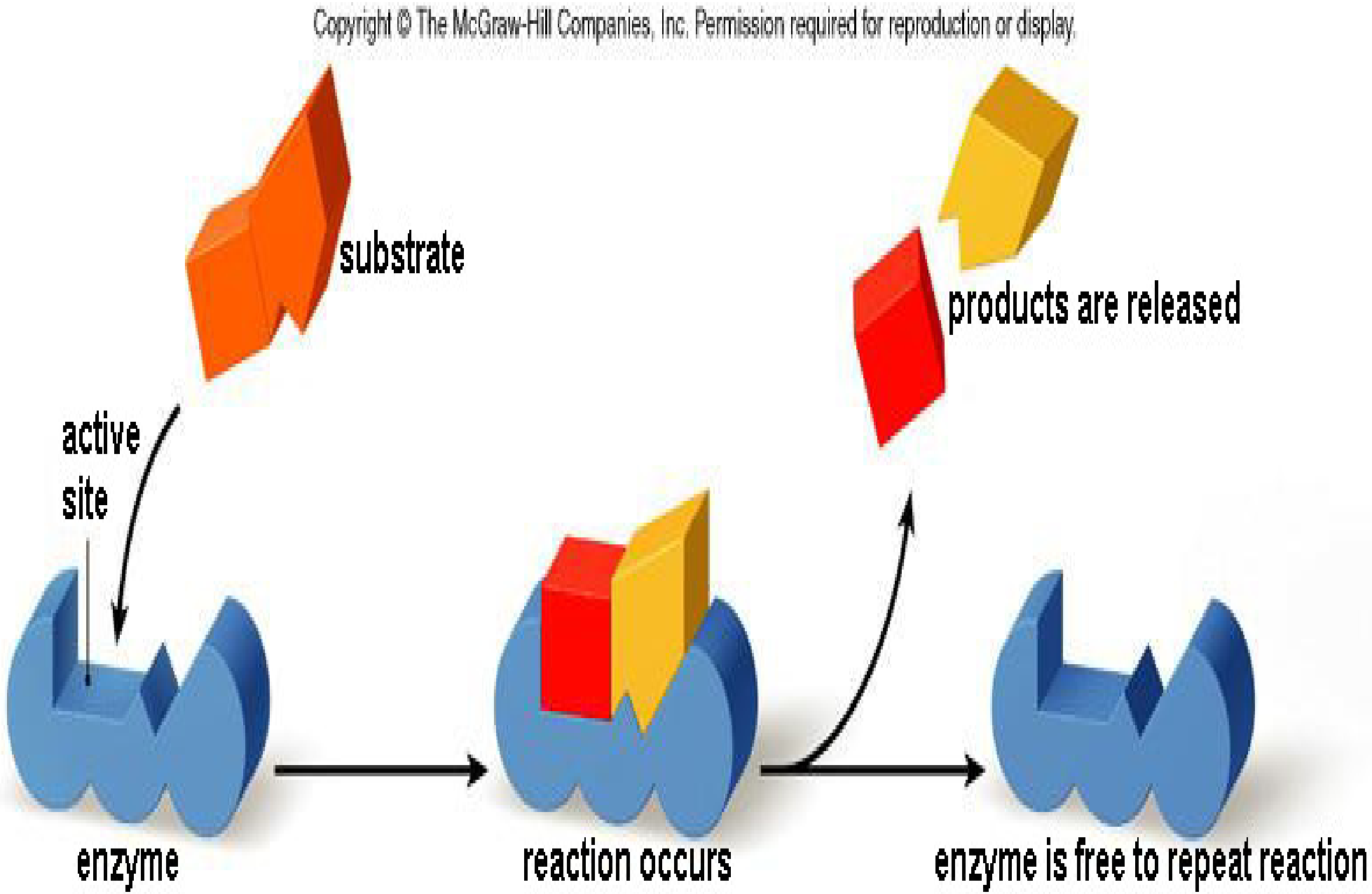
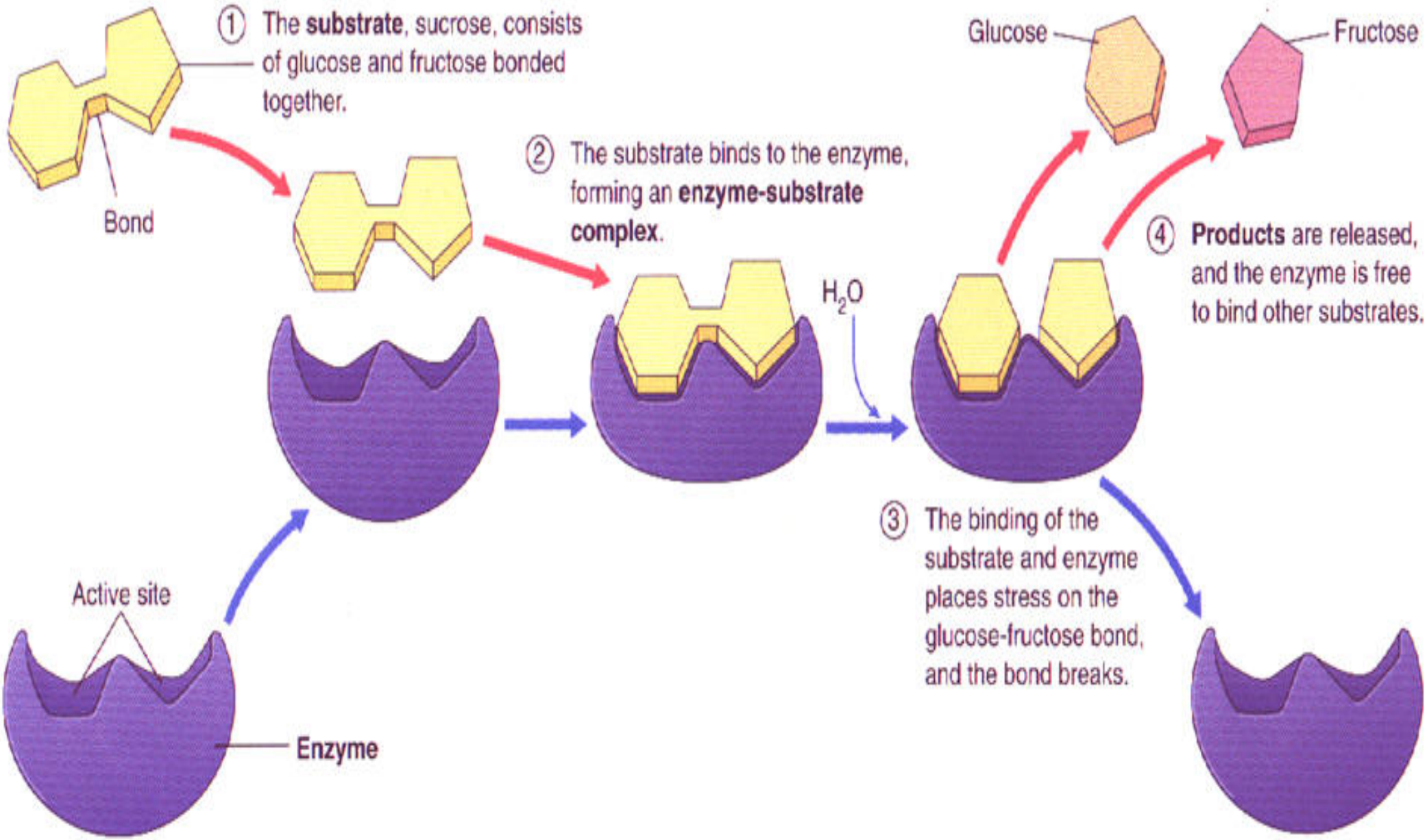
Enzyme in action (Catabolic)



http://legacy.hopkinsville.kctcs.edu/

Catabolic Specific example



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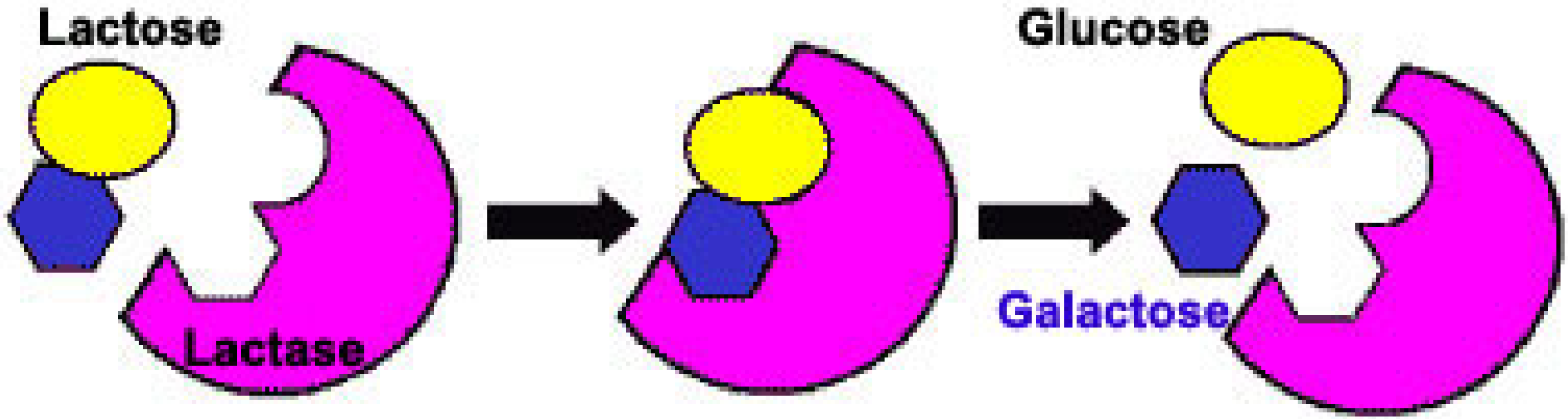
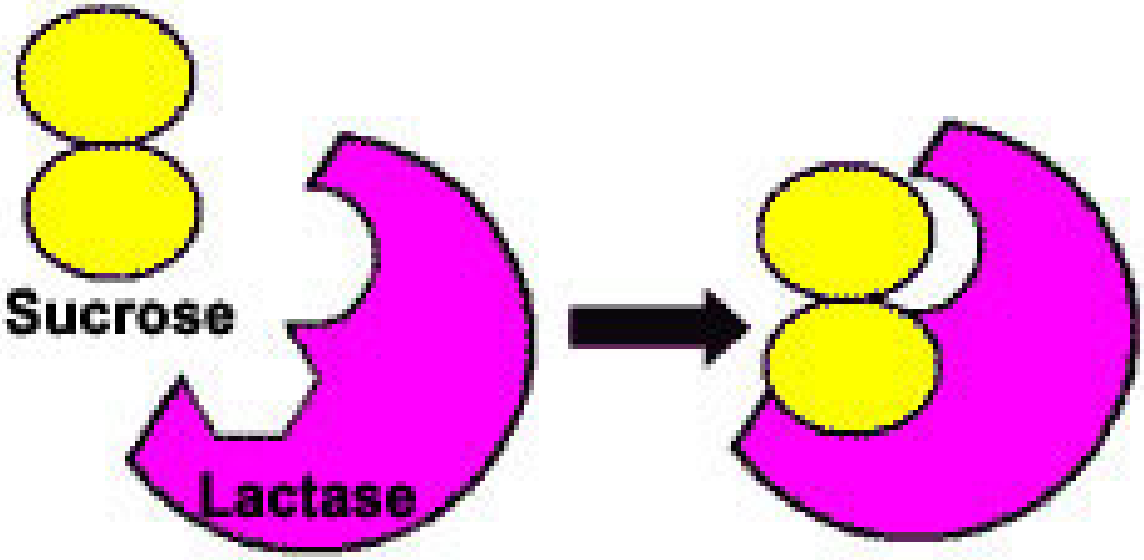
Enzyme Specificity





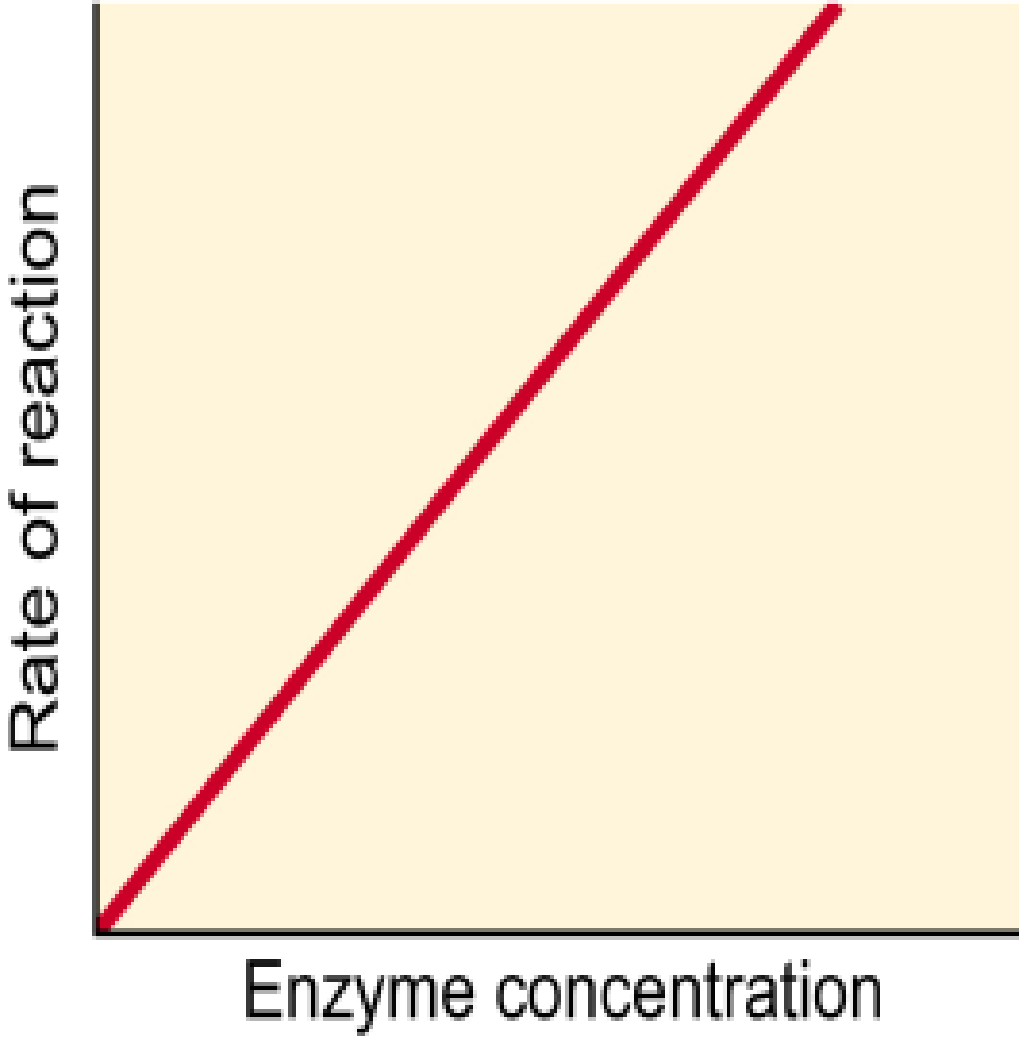
http://www.chemicalconnection.org.uk/

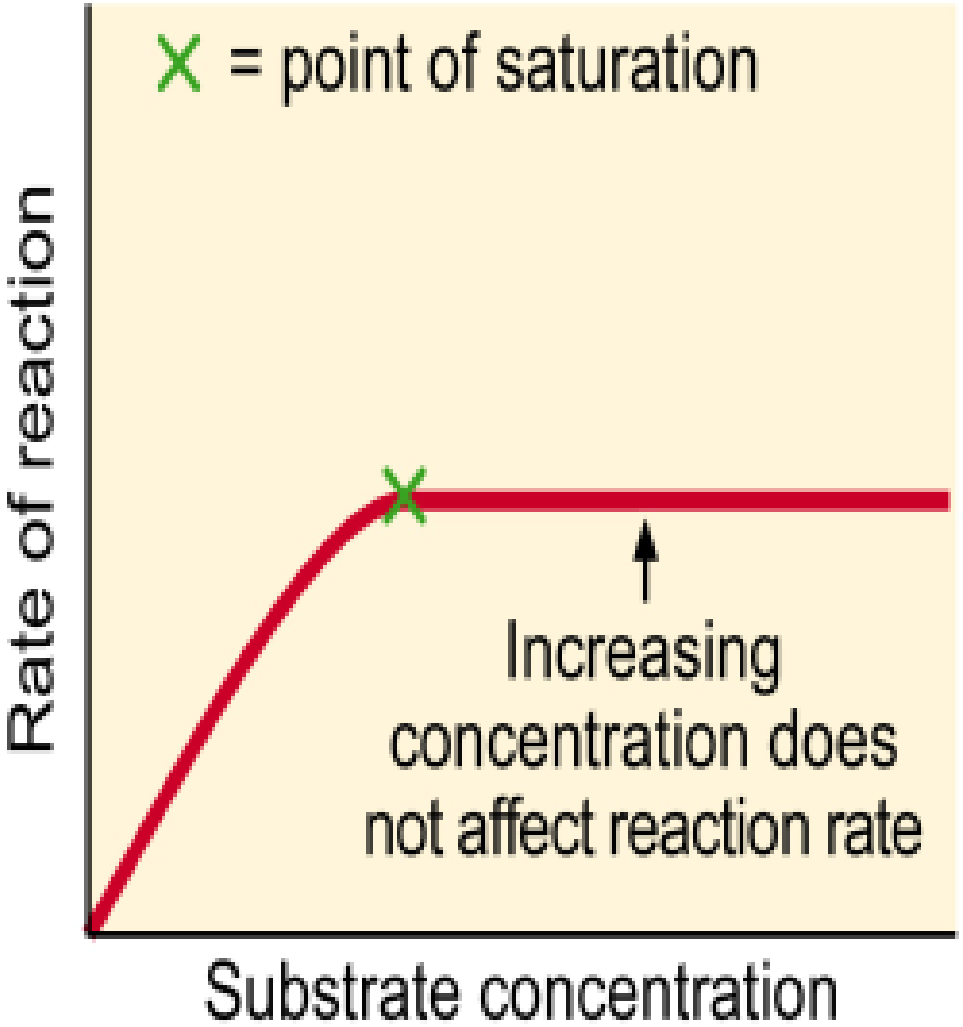
Factors affecting enzyme activity

- The rate of reaction of an enzyme-catalyzed reaction is affected by diverse factors.
- Some factor increase/decrease the rate, some inhibit/activate the enzyme action while some denature/destroy the enzyme
- **Enzyme/substrate concentration; temperature; pH; inhibitors; Enzyme cofactors (activators and coenzymes)**

Enzyme/ Substrate Concentration

- As the concentration of either is increased the rate of reaction increases
- For a given enzyme concentration, **the rate of reaction increases with increasing substrate concentration up to a point, above which any further increase in substrate concentration produces no significant change in reaction rate.** This is because the active sites of the enzyme molecules at any given moment are virtually **saturated with substrate**.





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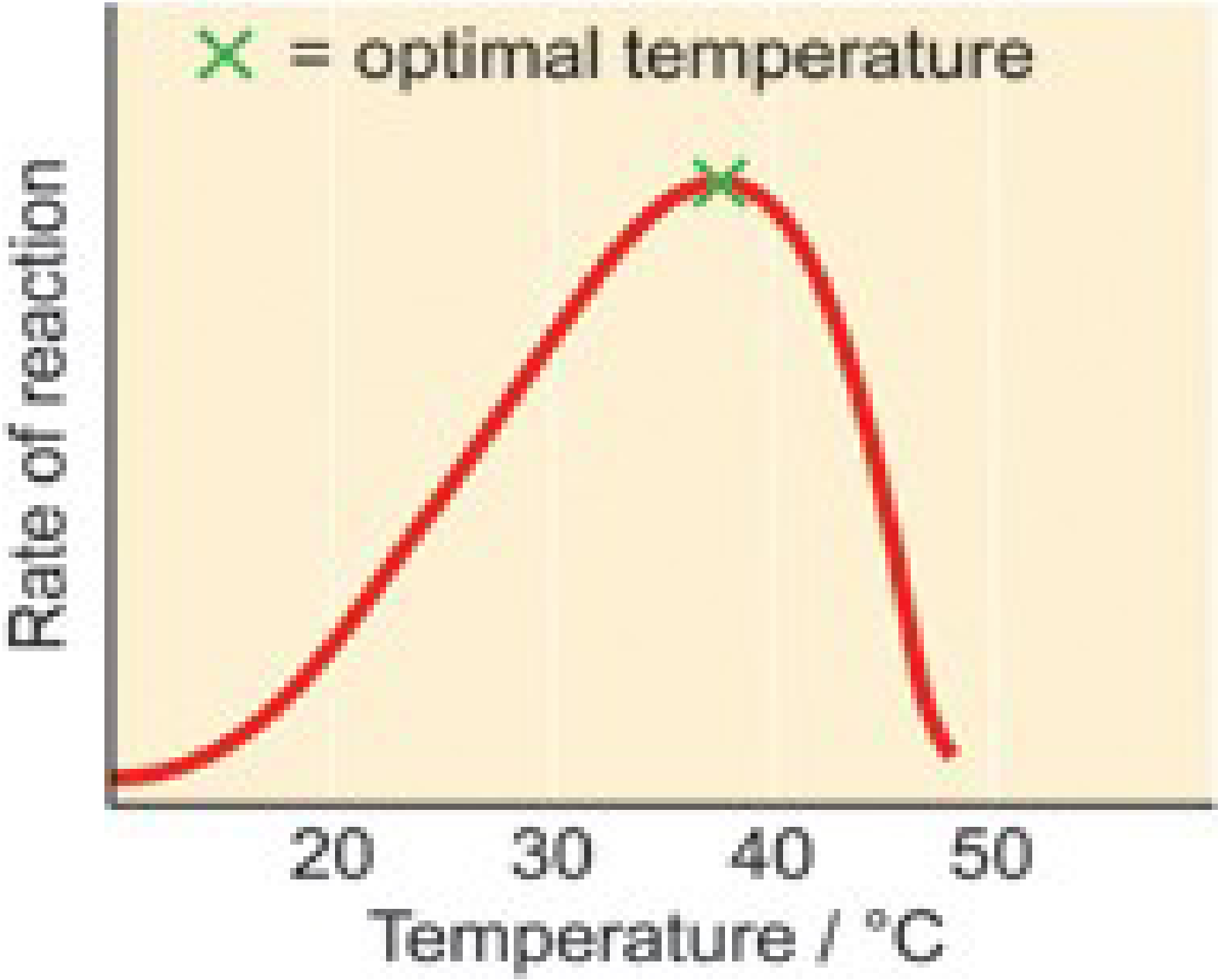
Temperature

- ***Heating increases molecular motion***, thus there is greater chance of the enzyme and substrate molecules bumping into each other and reacting.
- The temperature that promotes maximum activity is the **optimum temperature.**
- Above this temperature there is decrease in reaction rate because of **enzyme denaturation** but at lower temperature, the action is slowed and enzymes **inactivated** near/below freezing point.e.g. Frozen foods with innate enzymes.
- Human beings enzymes- 35°-40°C optimal; Hot spring bacteria enzymes- ≥ 70°C optimal.



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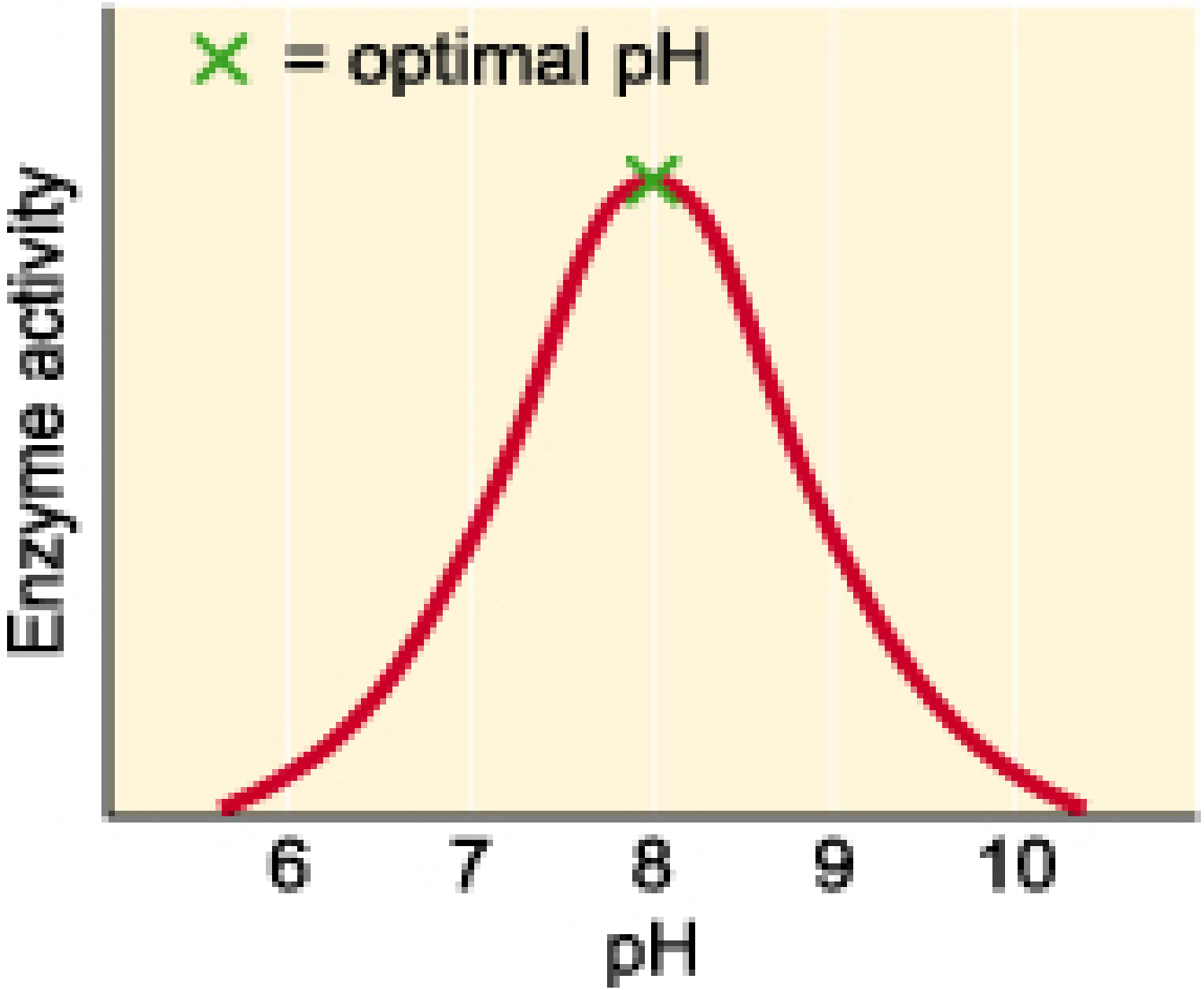
pH

- pH **affects the shapes of protein molecules** (including enzymes) as hydrogen bonds and sulphur bridges are broken or formed.
- Different enzymes function effectively at diverse narrow pH range. There is also an **optimum pH** for each enzyme above/below which enzyme activity diminishes, usually pH 6 to 8.

However, some enzymes such

- Pepsin (in stomach) operates at low pH 2-3 while
- Trypsin (in small intestine) operates at pH 8 to 9.



http://www.rsc.org/Education/Teachers/Resources/cfb/enzymes.htm

Inhibitors

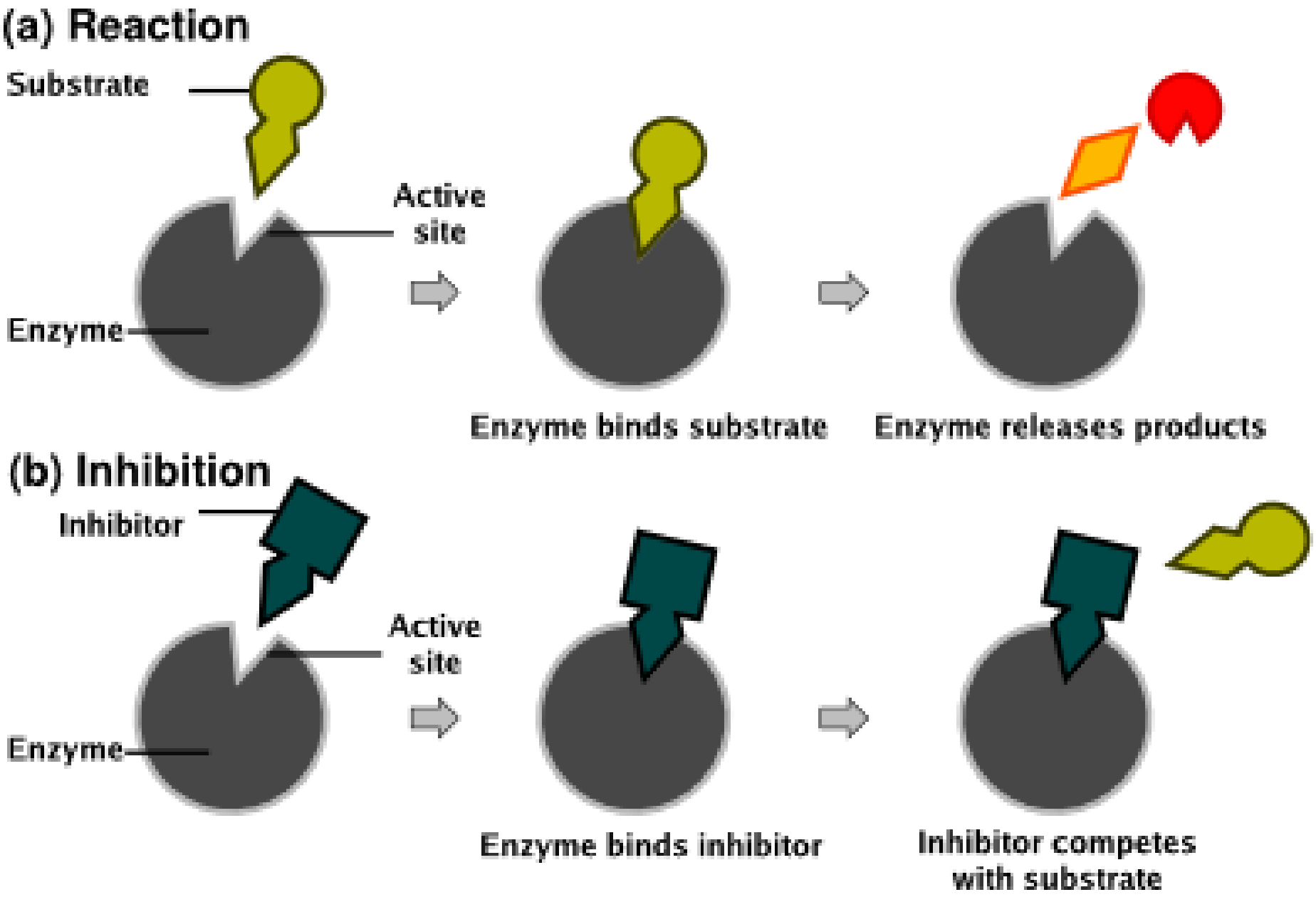
- These are substances/molecules that **stop enzymes** from working or **reduce their catalytic ability**.
- There are two main types of inhibition:

-**Competitive inhibition**

-**Non-competitive inhibition** (two types, viz: ***Reversible or irreversible***)

Competitive inhibition

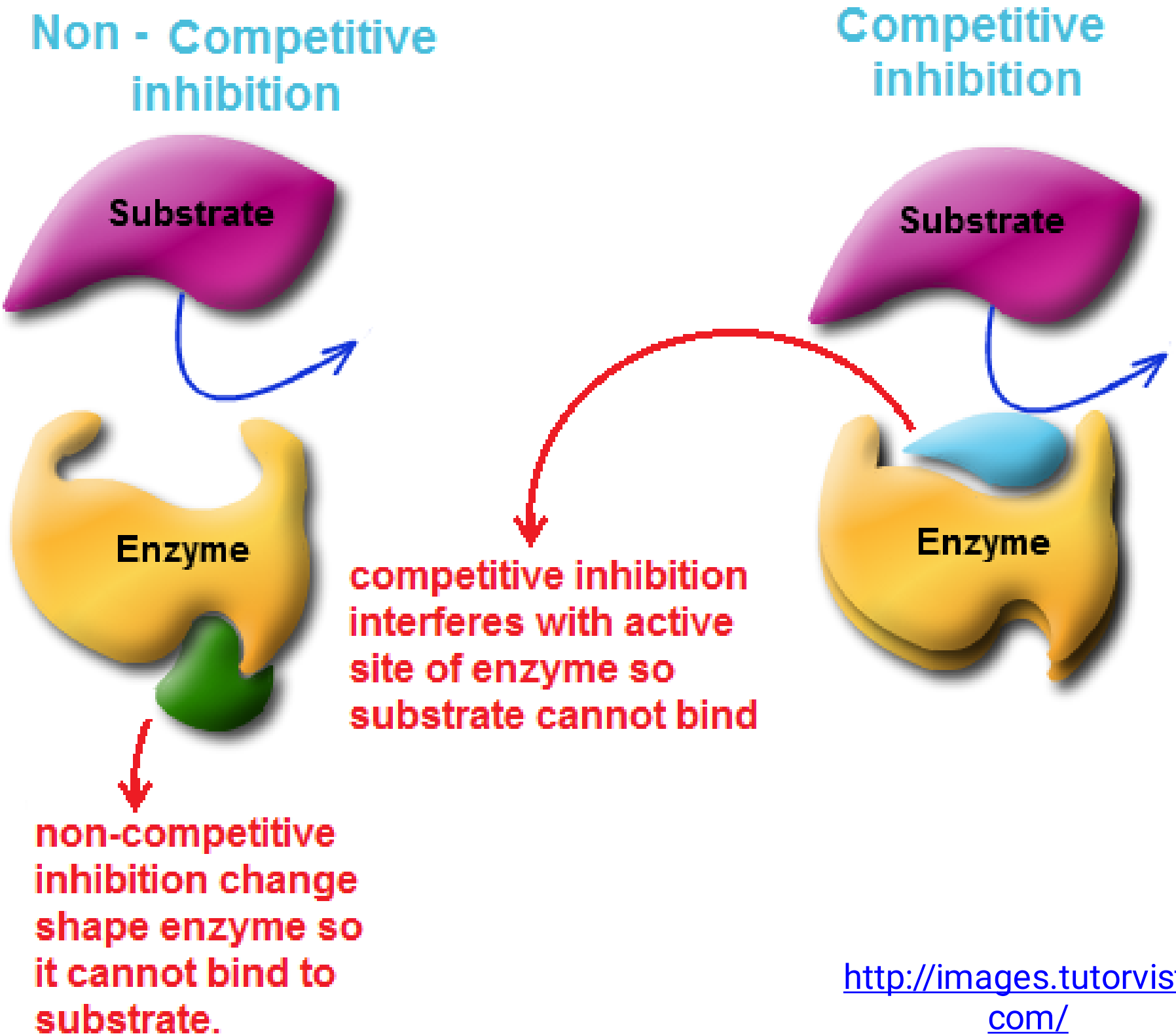
- This occur when a substance **has a structure which is sufficiently similar to that of the normal substrate** to be able to fit into the active site .
- It **prevents** the true substrate from entering the active site.
- The substrate and inhibitor hence **compete for the active site**.
- However, if the **substrate concentration is increased**, the rate of reaction increases.



http://gleesonbiology.pbworks.com/

Non-Competitive reversible inhibition

- Inhibitor has **no structural similarity to substrate.**
- **Inhibitor attach to another site on the enzyme** molecule other than the active site, **distorting the shape of the active site**.
- This makes it impossible for substrate to bind to the active site.
- However, **when the inhibitor leaves, the active site return to shape** and becomes usable to the substrate.
- Increase in **substrate concentration have no effect** unlike the competitive inhibition.



http://images.tutorvista.com/

Non-competitive irreversible inhibition

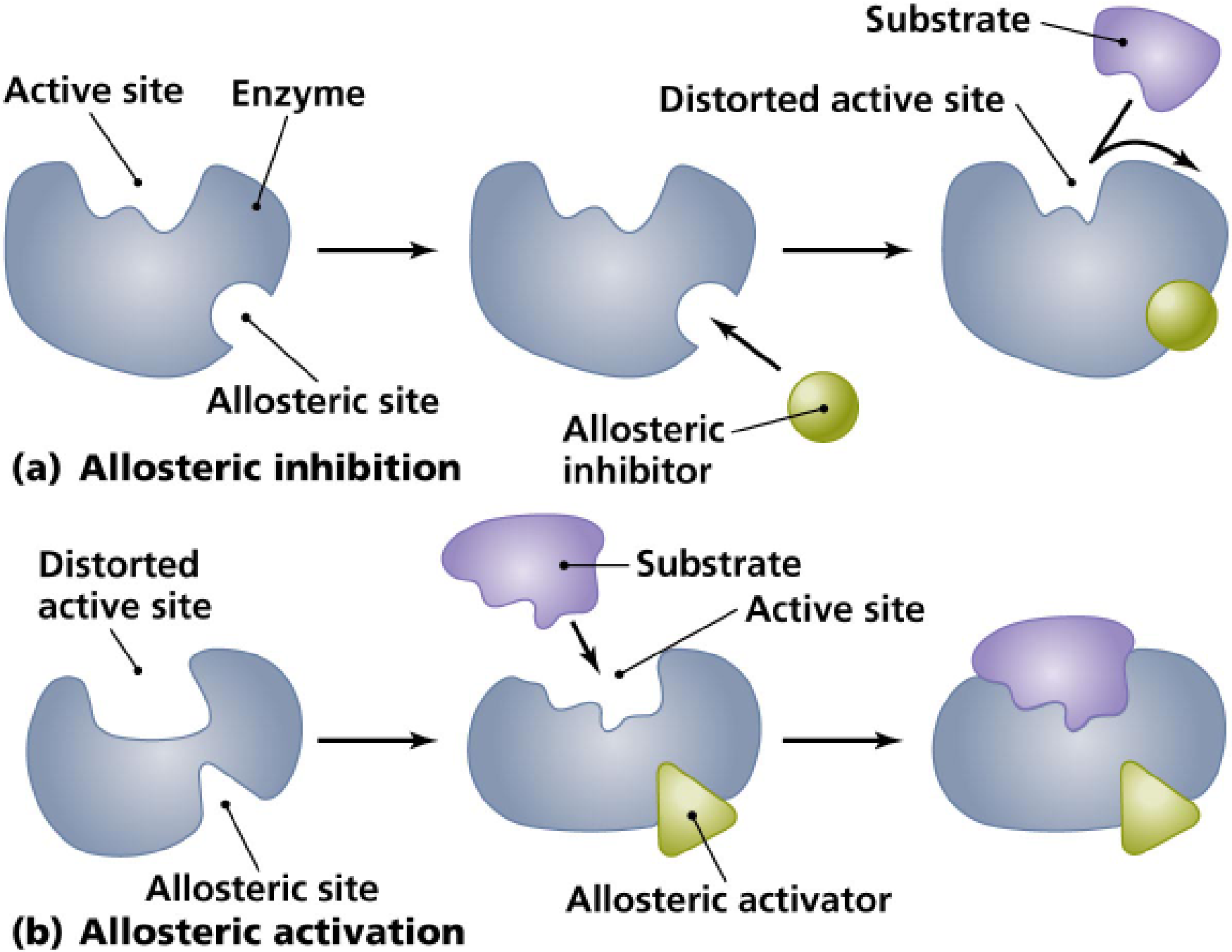
- Inhibitor combines with the enzyme permanently
- It changes the shape and structure of the enzyme irreversibly, hence inactivating the enzyme permanently.
- It occurs slowly than other inhibition but more devastating.
- Examples of such inhibitor include cyanide and mercury.

Enzyme cofactors

- These are non-protein part that enables certain enzymes to function.
- These could be **inorganic ions**, called **activators,** they bind to the enzyme/substrate complex and **makes it function more effectively**. E.g. Salivary amylase works better in the presence of chloride ions.
- They could be **organic molecules**, called **coenzymes. They loosely associate with the enzyme.** They **transfer chemical groups, atoms or electron from one enzyme to another.** Many of them are **vitamins or vitamin derivatives**

Allosteric enzymes

- Greek: *allos* “other” + *steros* “shape/form”
- Allosteric enzymes **change shape between active and inactive form** as a result of the binding of substrates at the active site, and of **regulatory molecules** at other sites.
- These other sites serves as chemical on/off switches.
- **If inhibitor binds to such site, it stabilises the inactive** form/shape hence it reduces the enzyme activity while a bound **activator stabilises the active** form/shape thereby increasing enzyme activity



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http://academic.pgcc.edu/~kroberts/

- Anabolic video
- http://www.youtube.com/watch?v=PlLzvT3spCQ Enzyme function and inhibition (with audio narration)

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